# Stick-together calculations

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Find a common final velocity using total momentum.

**SUCCESS CRITERIA - I can...**

1. Add the initial momenta with signs.
2. Use the combined mass after sticking.
3. Check the result against the direction of total momentum.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

Fragments of masses 3 and 5 kg separate from rest. The 3 kg piece moves at +10 m/s. Find the other velocity and total kinetic energy.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $v_f = (m_1 v_{1,i} + m_2 v_{2,i})/(m_1 + m_2)$.
Sticking collision in one dimension with negligible external impulse. $v_f$ is the shared final velocity.

**Worked reasoning:** A 2 kg cart at +3 m/s sticks to a 1 kg cart at rest. Total momentum 6 kg·m/s divided by total mass 3 kg gives final velocity +2 m/s.

**Your turn:** Mark the conserved quantity on a before-and-after sketch. Explain why masses are added in the final denominator.

| Before | After |
|---|---|
| | |

Label mass, velocity and momentum.

**Marzano check - circle:** ① ② ③

## 03 Read and interpret the model

Two carts stick. Identify which masses belong in the final momentum expression and why.

Before contact: add velocity arrows.




**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A 2 kg cart at +5 m/s sticks to a 3 kg cart at -1 m/s. Find final velocity and kinetic energy transformed. Organize the known quantities on the before-and-after model.

| Before | After |
|---|---|
| | |

Label objects and signed velocities.

**Marzano check - circle:** ① ② ③

# Stick-together calculations

**antocciphysics**



Continue down each column.

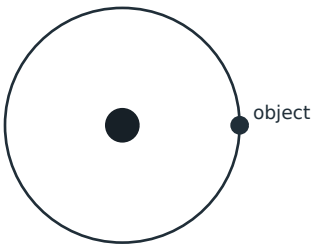
## 05 Explain your reasoning

A student averages the two starting velocities without considering mass. When would that happen to work?

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

At fixed mass and speed, circle radius becomes 4 times as large. Compare acceleration and net force using equations.
Label velocity and inward acceleration on the circular model.




**Marzano check - circle:** 1 2 3

## 07 Exit check

A 0.20 kg projectile at 30 m/s sticks to a 1.80 kg cart at rest. Find common speed and fraction of initial kinetic energy remaining. Organize the known quantities on the before-and-after model.

| Before | After |
| --- | --- |
| | |

Label objects and signed velocities.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

A 1 kg cart at +6 m/s sticks to a 2 kg cart at -3 m/s. Find final velocity. If the second velocity were -1 m/s instead, find the new result. Organize the known quantities on the before-and-after model.

| Before | After |
| --- | --- |
| | |

Label objects and signed velocities.

**Marzano check - circle:** 1 2 3